Answer Key - Health Sciences - Graduate Level Assessment 11 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) Adaptation; B) Control; C) Growth; D) Generativity; E) Recognition
   Note: Adaptation is one of McAdam's levels of personality because it refers to the ways in which individuals adjust their behavior and thinking in response to their environment, reflecting a key aspect of personality development in health sciences.

2. **Answer:** A
   Options: A) Determining what first causes the senescence in the immune system; B) The immune system theories do not explain the role of lifestyle in senescence; C) Immune system theories do not explain the role of genetic factors in senescence; D) Immune system theories are too complex to test effectively; E) The lack of information about senescence in general
   Note: The correct answer highlights the challenge of identifying the initial triggers or factors that lead to the decline in immune system function with age, which is a central issue in understanding the biological mechanisms of senescence.

3. **Answer:** A
   Options: A) cervical spinal nerves.; B) optometric nerves.; C) lumbar spinal nerves.; D) hypoglossal nerves.; E) olfactory nerves.
   Note: The correct answer is based on the fact that parasympathetic preganglionic axons originate from the brainstem and sacral spinal cord, but in the case of cervical spinal nerves, the vagus nerve, which carries parasympathetic fibers, exits the CNS in the cervical region, thus highlighting the connection between parasympathetic function and cervical spinal nerves.

4. **Answer:** C
   Options: A) Reptiles; B) Birds; C) Primates; D) Humans; E) Cattle
   Note: The correct answer is "Primates" because both Lassa and Ebola viruses are zoonotic pathogens that have been linked to specific primate species, highlighting the role of wildlife, particularly primates, in the emergence and transmission of these viruses in West Africa.

5. **Answer:** A
   Options: A) carcinoma.; B) kidney stones.; C) peptic ulcer.; D) appendicitis.; E) gastroenteritis.
   Note: Carcinoma is a leading cause for stoma formation because it often necessitates surgical intervention to remove cancerous tissue in the gastrointestinal tract, which may result in the creation of a stoma for waste elimination.

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: Testing accessory nerve function involves assessing the strength of the sternocleidomastoid and trapezius muscles, which requires resistance against head turning, rather than simply asking the patient to turn their head without resistance.

7. **Answer:** True
   Options: A) True; B) False
   Note: Frequent use of antibiotics can disrupt the balance of gut microbiota, which may enhance immune system regulation and reduce the incidence of allergic conditions.

8. **Answer:** False
   Options: A) True; B) False
   Note: MRI is not typically used for detecting neural tube defects because ultrasound is the preferred imaging modality during pregnancy for evaluating fetal development and identifying such anomalies.

9. **Answer:** True
   Options: A) True; B) False
   Note: The normal respiratory rate for an adult male falls within the range of 12 to 20 breaths per minute, making the specified range of 14 to 16 breaths per minute accurate and true.

10. **Answer:** True
    Options: A) True; B) False
    Note: Noroviruses are known to infect the gastrointestinal tract, leading to acute gastroenteritis characterized by symptoms such as projectile vomiting and diarrhea due to the virus's ability to rapidly replicate and disrupt the intestinal lining.

Long Form Response

11. **Answer:** The stomach lining contains several types of cells, including mucous cells, chief cells, and parietal cells, each contributing to the organ's digestive functions. Parietal cells are particularly significant as they secrete hydrochloric acid and intrinsic factor, but they also play a crucial role in the secretion of pepsinogen, the inactive precursor of the enzyme pepsin. When pepsinogen is released into the acidic environment of the stomach, it is activated to pepsin, which is essential for the digestion of proteins into smaller peptides. This activation process underscores the importance of parietal cells in facilitating protein digestion, ultimately aiding nutrient absorption and overall digestive efficiency. Thus, the function of parietal cells and their role in pepsinogen secretion is vital for the digestive process within the stomach.
    Note: correct because it accurately identifies the types of cells in the stomach lining, emphasizes the essential role of parietal cells in secreting hydrochloric acid that activates pepsinogen to pepsin, and highlights the significance of this process in digestion.

12. **Answer:** Mixed mitral valve disease with a predominance of mitral regurgitation presents several clinical features that are critical for diagnosis. The tapping apex beat often indicates left ventricular hypertrophy due to volume overload, reflecting the chronic nature of mitral regurgitation. A loud first heart sound is significant as it suggests increased pressure in the left atrium and highlights the severity of the regurgitation. The presence of a pan-systolic murmur, best heard at the apex and often radiating to the left axilla, signifies continuous backflow of blood into the atrium during ventricular contraction. Additionally, a long mid-diastolic murmur can indicate significant atrial volume overload, resulting from prolonged diastolic filling due to the regurgitant flow. Together, these clinical signs help create a comprehensive picture of mixed mitral valve disease, guiding clinicians in both diagnosis and management strategies.

    Note: The answer correctly identifies and explains the significance of key clinical features of mixed mitral valve disease with a predominance of mitral regurgitation, such as the tapping apex beat indicating left ventricular hypertrophy, a loud first heart sound indicating elevated left atrial pressure, and a pan-systolic murmur reflecting the severity of regurgitation, all of which are interrelated and crucial for accurate diagnosis.

End of Answer Key